CAT405 Intelligent Computing Major Project

System Requirement and Design (SRD) Presentation

Project Code : PenangLens

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163634

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Examiner : Dr. Nur Intan Raihana Ruhaiyem

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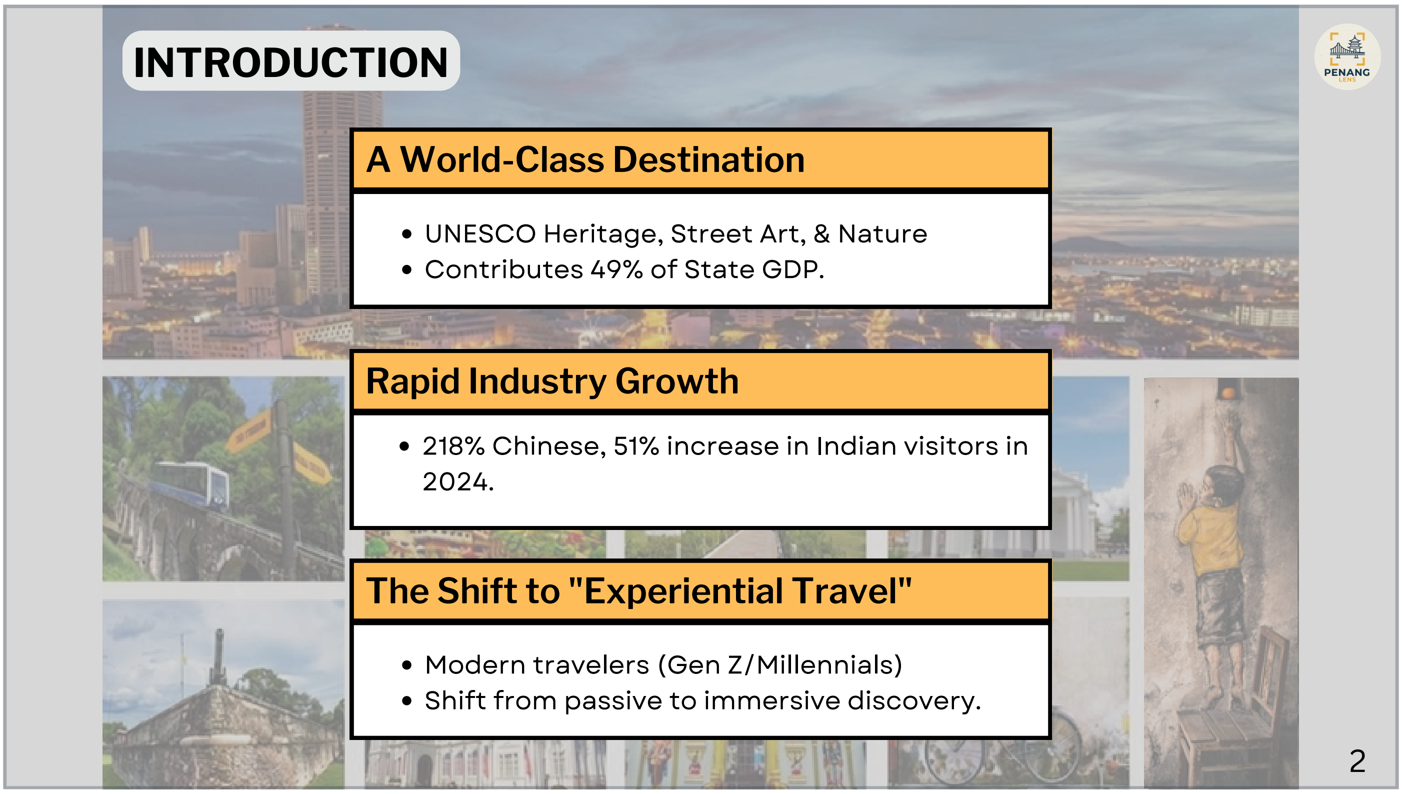
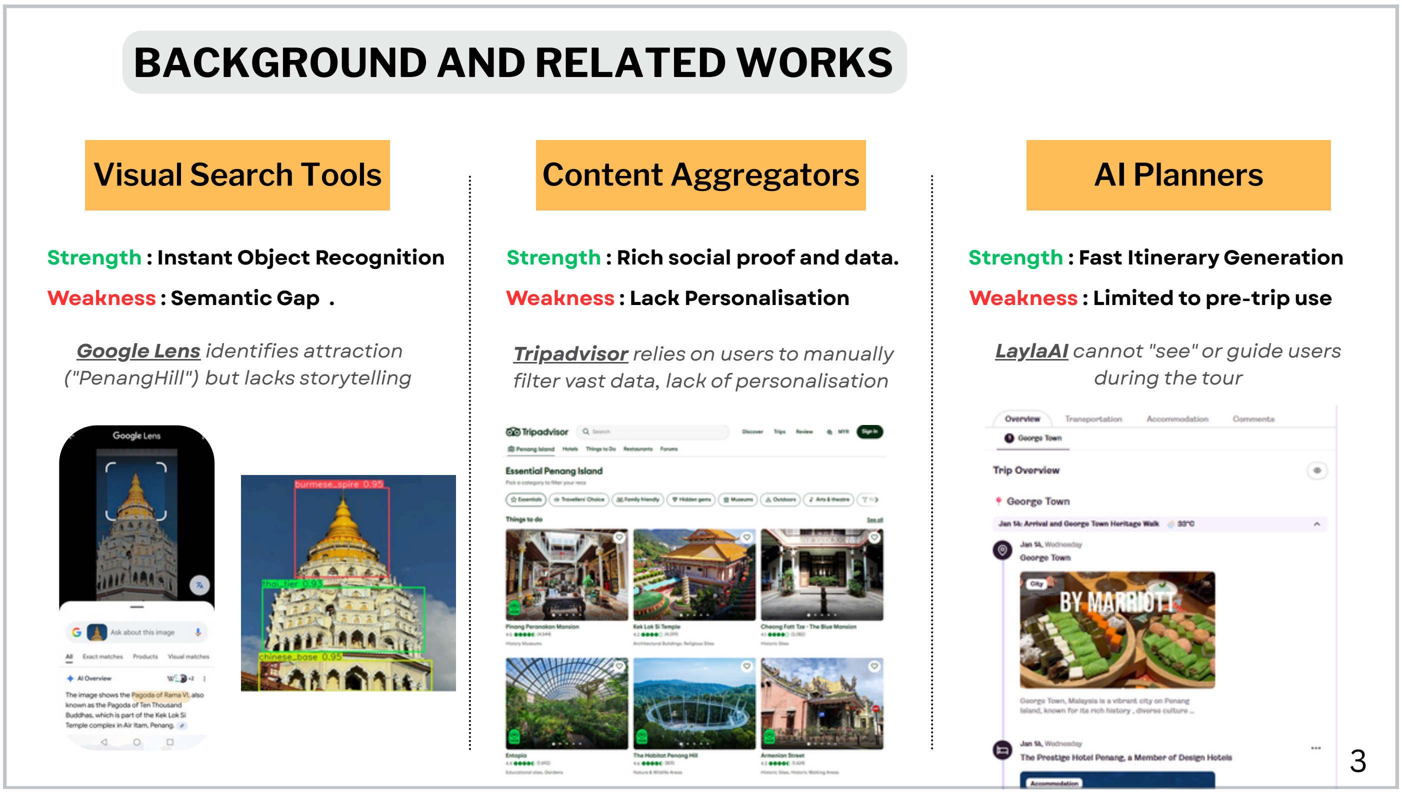
Hello, Dr. Gan and Dr. Intan. My name is Ting Juin, and I will be presenting the (SRD) for PenangLens.



Penang is a world class destination which become even famous since George Town became a UNESCO World Heritage Site. As stated in the Penang Tourism Master Plan, the service sector contributes a total of 49% of the state's GDP, with tourism being one of the biggest contributor.

Additionally the tourism industry is currently experiencing a massive boom.

Besides we are seeing a shift toward 'Experiential Travel. where Modern travelers are no longer satisfied with just 'checking in' at a landmark but they want to know more about the story and details about what they are visiting.



So what are the existing tools exist in tourism market?

First are Visual Search tools like Google Lens. While they offer great instant recognition, they suffer from a 'Semantic Gap.' For instance, Google Lens can identify a landmark but cannot explain more details or the story behind.

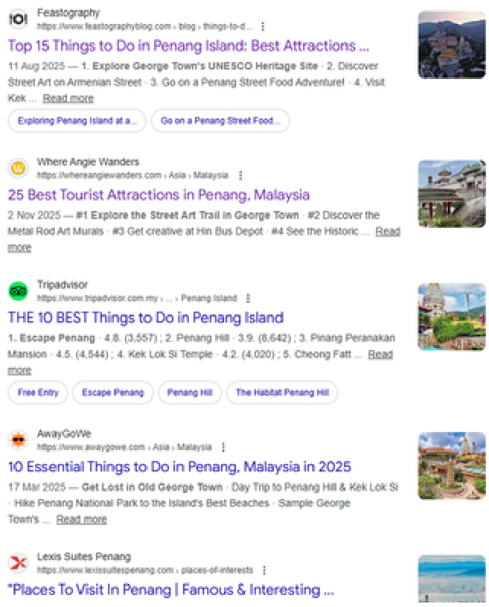
Second are Content Aggregators like TripAdvisor. Although these tools has rich social proof and data, but they lack of personalisation,.

Third are AI Planners like Layla AI. This is excellent for generating fast itineraries but only limited to pre-trip use and stop being useful at the moment when user starts travelling

PROBLEM STATEMENT

1 **Lack of Personalization**

- Static "Top 10" lists treat all tourists the same
- Irrelevant suggestions; system fails to capture specific user interests (History vs. Nature).


Feastography
https://www.feastographyblog.com › blog › things-to-d...
Top 15 Things to Do in Penang Island: Best Attractions ...
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Where Angie Wanders
https://whereangiewanders.com › Asia › Malaysia
25 Best Tourist Attractions in Penang, Malaysia
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Tripadvisor
https://www.tripadvisor.com.my › ... › Penang Island
THE 10 BEST Things to Do in Penang Island
1. Escape Penang · 4.8. (3,557) ; 2. Penang Hill · 3.9. (8,642) ; 3. Pinang Peranakan Mansion · 4.5. (4,544) ; 4. Kek Lok Si Temple · 4.2. (4,020) ; 5. Cheong Fatt ... Read more
Free Entry | Escape Penang | Penang Hill | The Habitat Penang Hill

AwayGoWe
https://www.awaygowe.com › Asia › Malaysia
10 Essential Things to Do in Penang, Malaysia in 2025
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Lexis Suites Penang
https://www.lexissuitespenang.com › places-of-interests
"Places To Visit In Penang | Famous & Interesting ...

Gen Z Traveler ,
History Enthusiast.

4

Now let’s look at the problem statement through Sarah's perspective, a history fan visiting Penang

First, she found many reccomendation online but they are always generic, showing the same top 10 lists similar to other tourists, which lacks personalization.

PROBLEM STATEMENT

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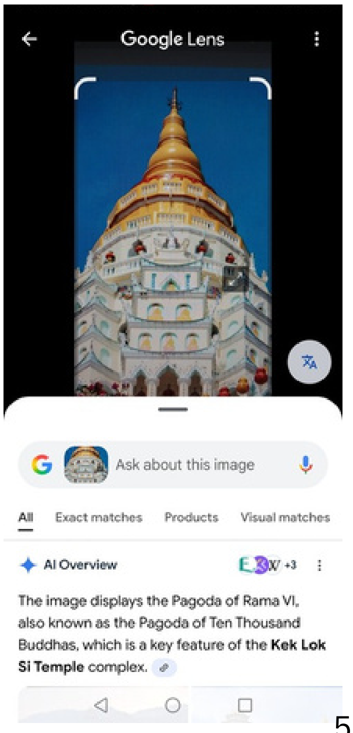
- Visual Tools provide Labels ("Kek Lok Si") but miss Interpretation.
- Superficial "check-in photography" without understanding architectural details or history.

Gen Z Traveler , History Enthusiast.



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Then , tools like Google Lens do not provide sufficient explanation when scanning an image. For example, it identifies this landmark as Kek Lok Si but stops there without providing further details.

PROBLEM STATEMENT

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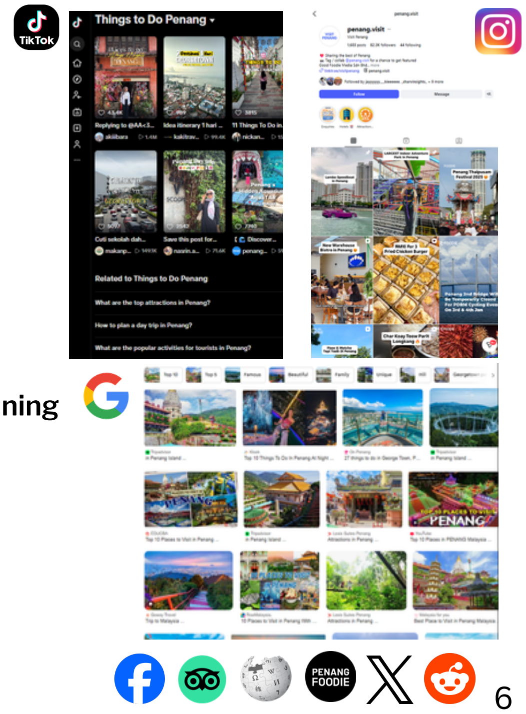
- Superficial "check-in photography" without understanding architectural details or history.

3 **Information Overload & Inefficient Planning**

- "Paradox of Choice" from scattered data sources
- Struggle to answer real-time queries (e.g., "What to do in 3 hours near Armenian Street?").

Gen Z Traveler , History Enthusiast.



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Third, planning is chaotic due to information overload, there are so many available sources from the web or social media that she does not know where to start.

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3 Information Overload & Inefficient Planning

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- Struggle to answer real-time queries (e.g., "What to do in 3 hours near Armenian Street?").

4 Scalability of Expert Guidance

- Human guides are the gold standard but unscalable.
- Impossible to provide expert-led storytelling to millions of tourists manually.

Gen Z Traveler , History Enthusiast.

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Finally, with millions of visitors yearly, Sarah can't easily get a private guide and is left exploring alone.

OBJECTIVES

1 Personalized Tourist Experience Module

- Develop a secure personalization engine.
- Generate User Preference Vectors (embeddings) to mathematically match user interests

2 Immersive AI Visual Discovery Module

- Construct a vision pipeline (DINOv2 + YOLO11) to identify landmarks and architectural details.
- Leverage Generative AI to create engaging, context-aware stories.

3 Intelligent Itinerary & Travel Companion

- Implement a proactive AI Agent (Chat interface).
- Generate feasible travel plans and provide an interactive map for real-time discovery.

4 Smart Content Curation & Management

- Create an admin dashboard for content verification (Human-in-the-Loop).
- Monitor AI performance and manage hierarchical heritage data.

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So with these problem , we derived the 4 objectives that our system should achieve

Firstly is to develop a Personalized Tourist Experience Module that provides
a mobile application with personalization engine to create a tailored tourism experiences.

Secondly is to construct an Immersive AI Visual Discovery Module that accurately identifies
landmarks and architecture details from user photos and generate engaging stories

Next is to implement an Intelligent Itinerary & Travel Companion Module that functions as an AI agent to generate customized travel plans.

Finally is to create a Smart Content Curation & Management Module that
provides tools for full content management for admins, and allows for the review of AI performance via user feedback

With that objecitves introducing PENANGLENS, an ai powered intelligent travel companion for tourists in penang

SDG ALLIGNMENT

Goal 8: Decent Work and Economic Growth

8.9 By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products.

1. Promote sustainable tourism & local culture.
2. Captures share of "Experiential Tourism" market
3. Increases engagement & length of stay, boosting local economy

Goal 11: Sustainable Cities & Communities

11.4 Strengthen efforts to protect and safeguard the world's cultural and natural heritage.

1. Protect & Promote Cultural Heritage
2. Innovates digital dissemination & cultural education.
3. Deepens cultural awareness

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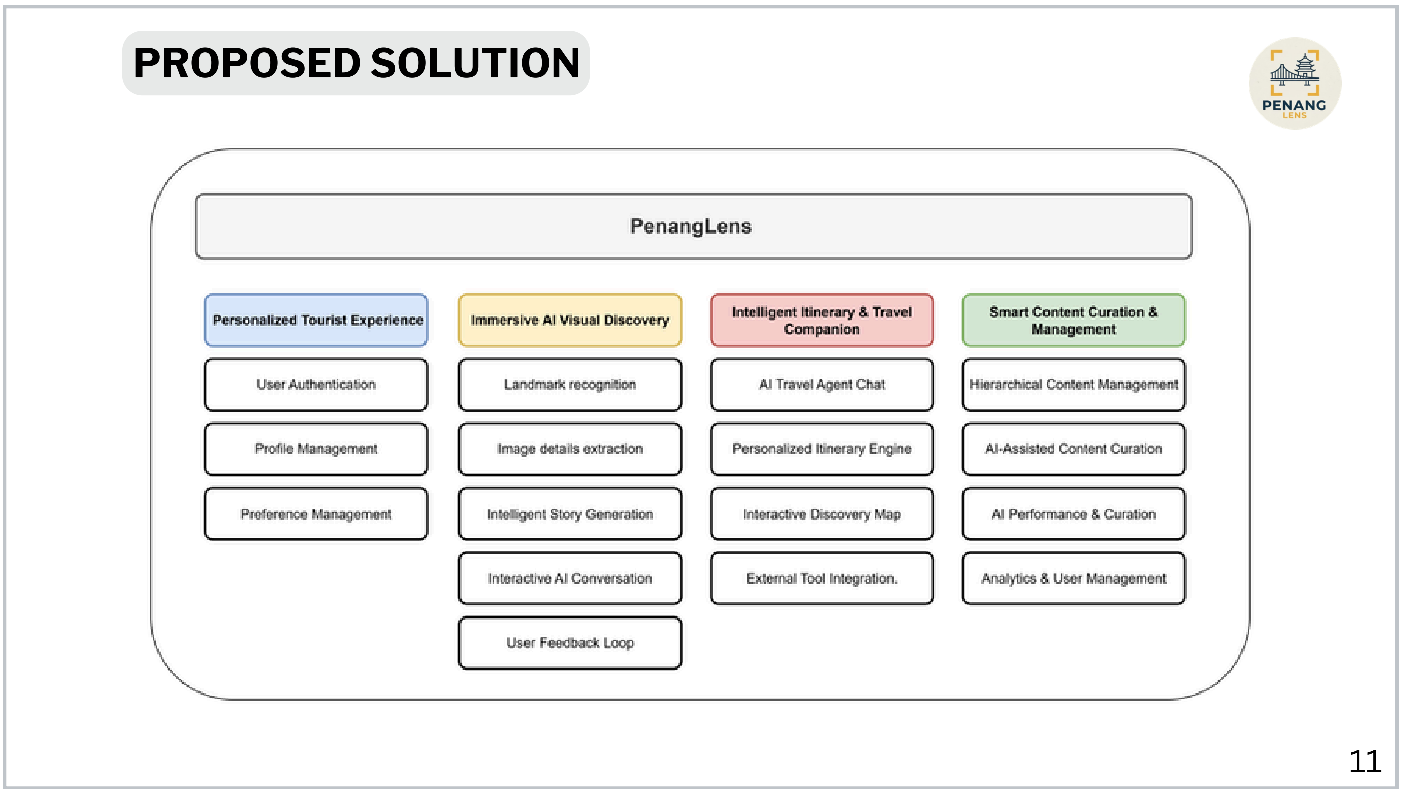
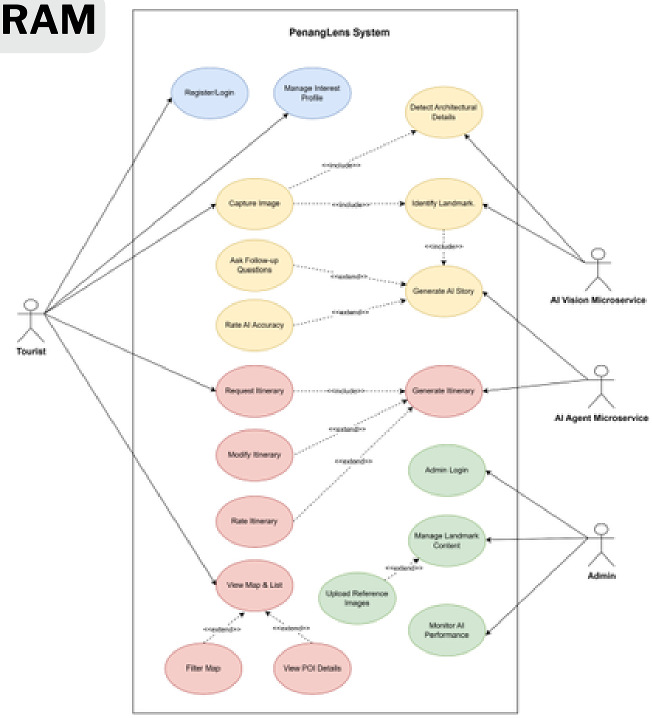
Besides that PenangLens is also aligned with SDG 8: Decent Work and Economic Growth which aims to promote sustainable tourism that boosts the local economy.

It also aligns with SDG 11: Sustainable Cities and Communities which focuses on deepening cultural awareness to protect the world’s heritage.



Showing here is our overall module diagram, PenangLens consists of four main modules with four core functions that we will explain later on

USE CASE DIAGRAM



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The two main actors in the system are tourists and admin.

In penanglens, Tourist utilizes the mobile application for visual discovery and intelligent itinerary planning.

While admin plays a crucial role in managing content and performing data verification to maintain the AI accuracy

now lets us move on to a more detailed explanation by module

MODULE 1 - PERSONALIZED TOURIST EXPERIENCE

1 Beyond Keywords

We don't just use simple tags.

2 Vector Embeddings

- User interests (e.g., "Street Art") and the location descriptions are converted into high-dimensional vectors
- Uses ***Google text-embedding-004***, 768-dimensional vectors

3 Cosine Similarity

Mathematically calculates the distance between User Vectors and Location Vectors to rank recommendations.

gemini-embedding-001

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So in Module 1 the personalised tourist experience module we focus on building the personalisation engine

we do not just use simple keyword matching to generate the user preference.

To solve this, we use Vector Embeddings. As shown in the flow chart, we first take the "User Interest Tags" and the "Location Descriptions" and embed them into a high dimension vectors, then we will be calculating the cosine similarity between the vectors and finally generate the personalised list to the user.

The embedding model use here will be the gemini-embedding-001 model , allowing the system to understand the "meaning" of a user's preference rather than just matching words.

```
locations = [
    {
        "name": "Kek Lok Si Temple",
        "description": "The largest Buddhist temple in Malaysia, featuring a seven-tiered pagoda, beautiful gardens, and a giant bron
    },
    {
        "name": "George Town Street Art",
        "description": "A collection of famous murals and wire sculptures scattered throughout the UNESCO World Heritage zone. Perfect
    },
    {
        "name": "Gurney Drive Hawker Centre",
        "description": "A famous seafront food court offering the best of Penang's street food. A paradise for foodies to try Char Ko
    },
    {
        "name": "The Habitat Penang Hill",
        "description": "A world-class rainforest discovery centre sitting on the fringes of a 130-million-year-old virgin rainforest.
    },
    {
        "name": "Escape Theme Park",
        "description": "An outdoor adventure theme park with water slides, obstacle courses, and zip lines. Designed for thrill-seeker
    }
]
```

```
👤 User Selected Tags:
   ['History', 'Temples', 'Street Art', 'Local Food']
🔤 Input to Model:
   "History, Temples, Street Art, Local Food"

ℹ️ Embedding User Profile...
ℹ️ Embedding 5 Locations and Calculating Similarity...

🎯 Personalization Results (Ranked by Similarity):
------------------------------------------------------------
LOCATION                     | SCORE      | MATCH REASON
------------------------------------------------------------
George Town Street Art       | 0.6632     | Good Match
Gurney Drive Hawker Centre   | 0.6211     | Good Match
Kek Lok Si Temple            | 0.6132     | Good Match
The Habitat Penang Hill      | 0.5648     | Low Relevance
Escape Theme Park            | 0.5423     | Low Relevance
```

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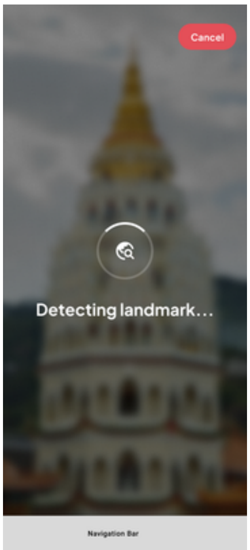
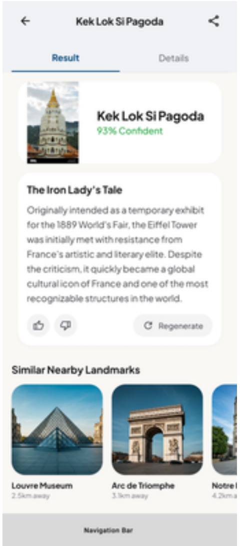
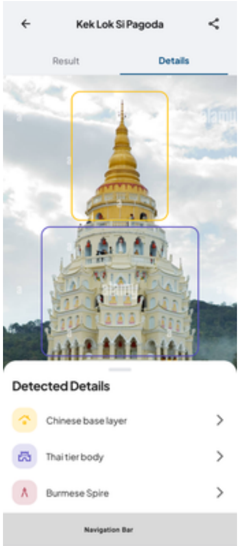
As shown in this quick prototype, the system calculates the semantic similarity between the user's interest tags and the location descriptions.

so at here George Town Street Art achieved the highest ranking because the description effectively consists of multiple user interests—specifically 'History' and 'Street Art'—simultaneously, resulting in a stronger vector alignment.

MODULE 2 - IMMERSIVE AI VISUAL DISCOVERY







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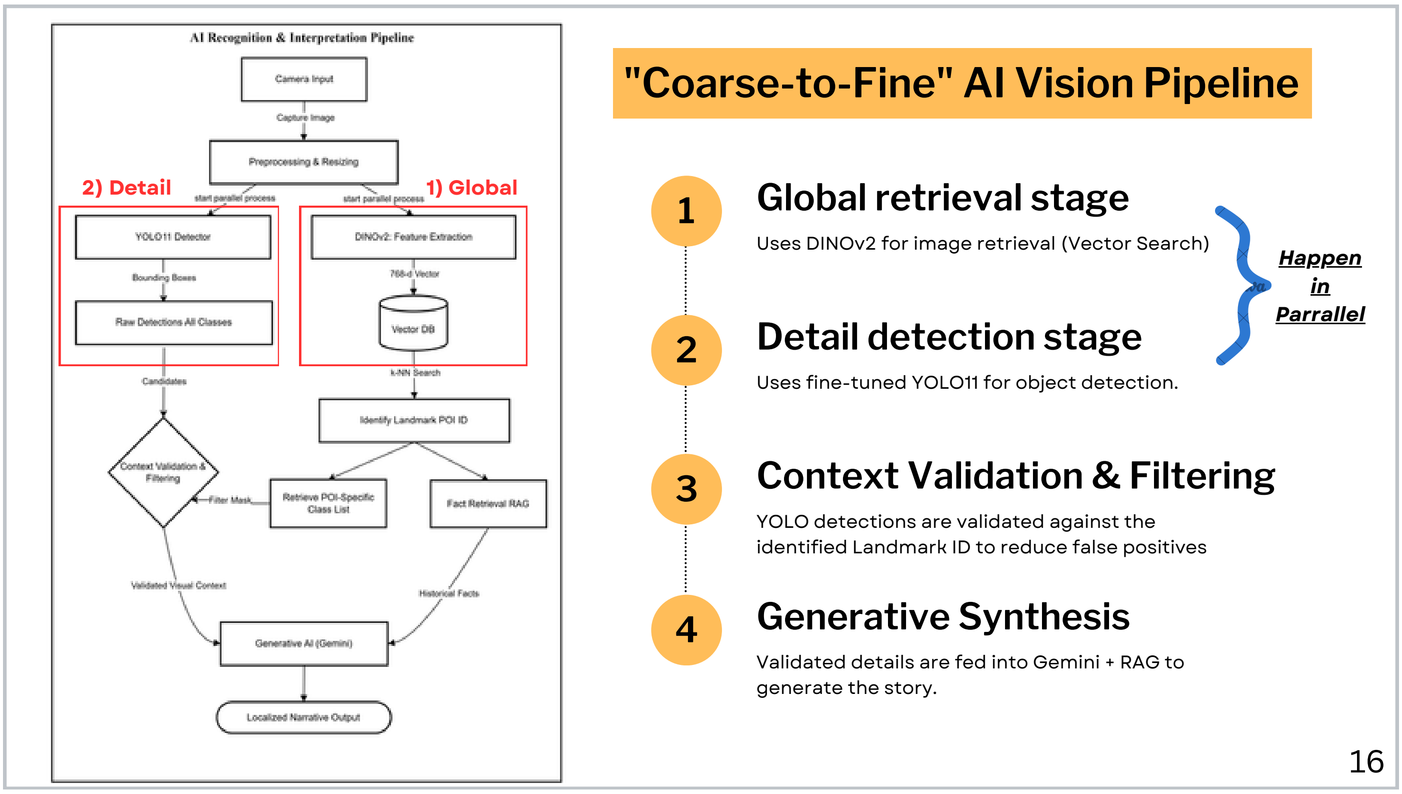
In Module 2, let's look at how the AI visual discovery works in PenangLens. First, a user takes a photo of a landmark. The system then identifies the landmark and provides an initial explanation. By navigating to the 'Details' tab, the user can see the image overlaid with bounding boxes that highlight specific architectural features. From there, the user receives more detailed stories tailored to those specific elements



So To achieve that level of detail, we designed a "Coarse-to-Fine" AI Vision Pipeline.

Lets take a look at the flow chart, when u user input an image, we first resize the raw image to 512 pixels which is the resolution we trained our model on

Then, the task is split into to steps which is the Global retrieval stage and detail detection stage

1. In the global retrieval stage, we use DINOv2 to perform a vector search and instantly identify the landmark's identity.

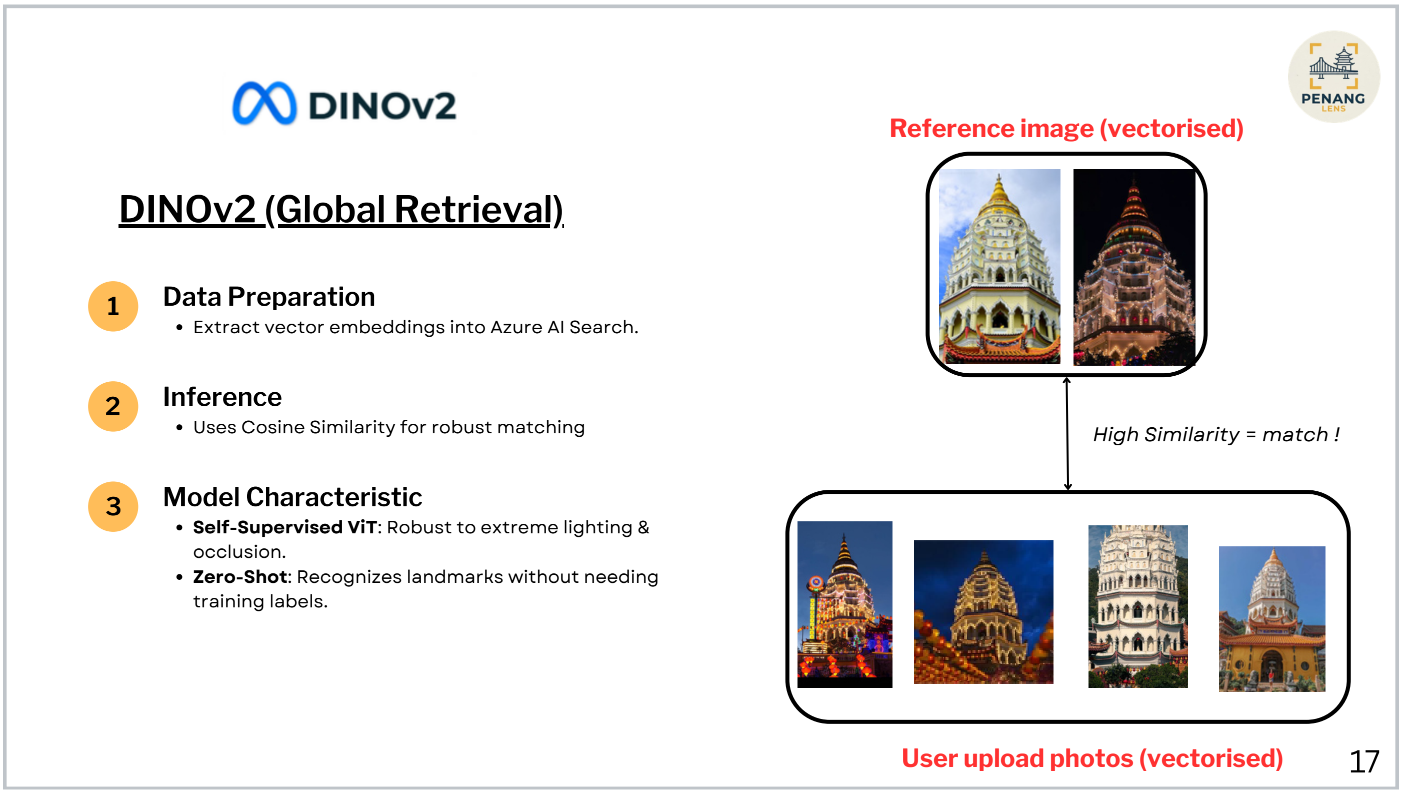
2. Simultaneously, in the detail detection stage, we run the YOLO11 detector to scan for specific architectural features.

And then when both steps above is done, will move on to the context validation & filtering step where We will validate the raw YOLO detections using the identified landmark.

For example, if the system identifies it as a kek lok si , it will filter out any unrelated detections like cable car, lampost or mural or even a tree

Next, Instead of relying solely on an LLM's general memory, which might lead to hallucinations, we use RAG and query our curated Vector Database for verified historical facts

With that we can generate a detailed story for the user.



Now, let's talk about the two models we will be using.

The first is DINOv2, which is used to convert reference images into high-dimensional feature vecto rs.

As in the example, we use dinov2 to vectorised the reference image and image uploaded by user, then by using cosine similarity, we are able to identify the uploaded photo's landmark.

So for data preparation, we only need to prepare reference image, vectorise it using DinoV2 and store to the azure ai search.

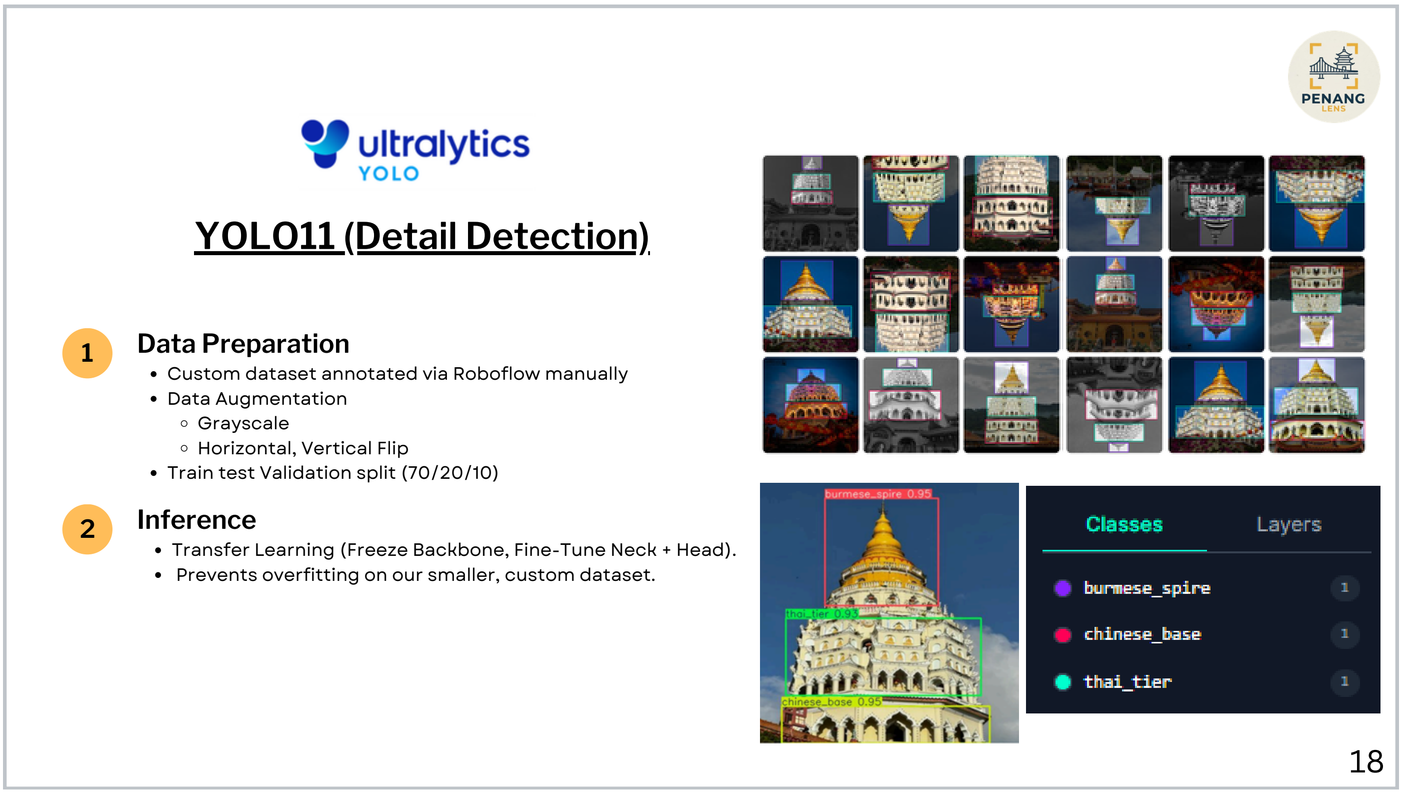
The key reason we chose DINOv2 is

First it is a self-supervised Vision Transformer, Self-supervised learning in DINO means the model learns visual patterns by comparing different views of the same image during training, without using any manually labeled data and. Hence robust to real-world challenges like extreme lighting or people blocking the view.

Secondly the zero-shot capability is allows us to index new landmarks instantly without needing to manually label thousands of training photos.

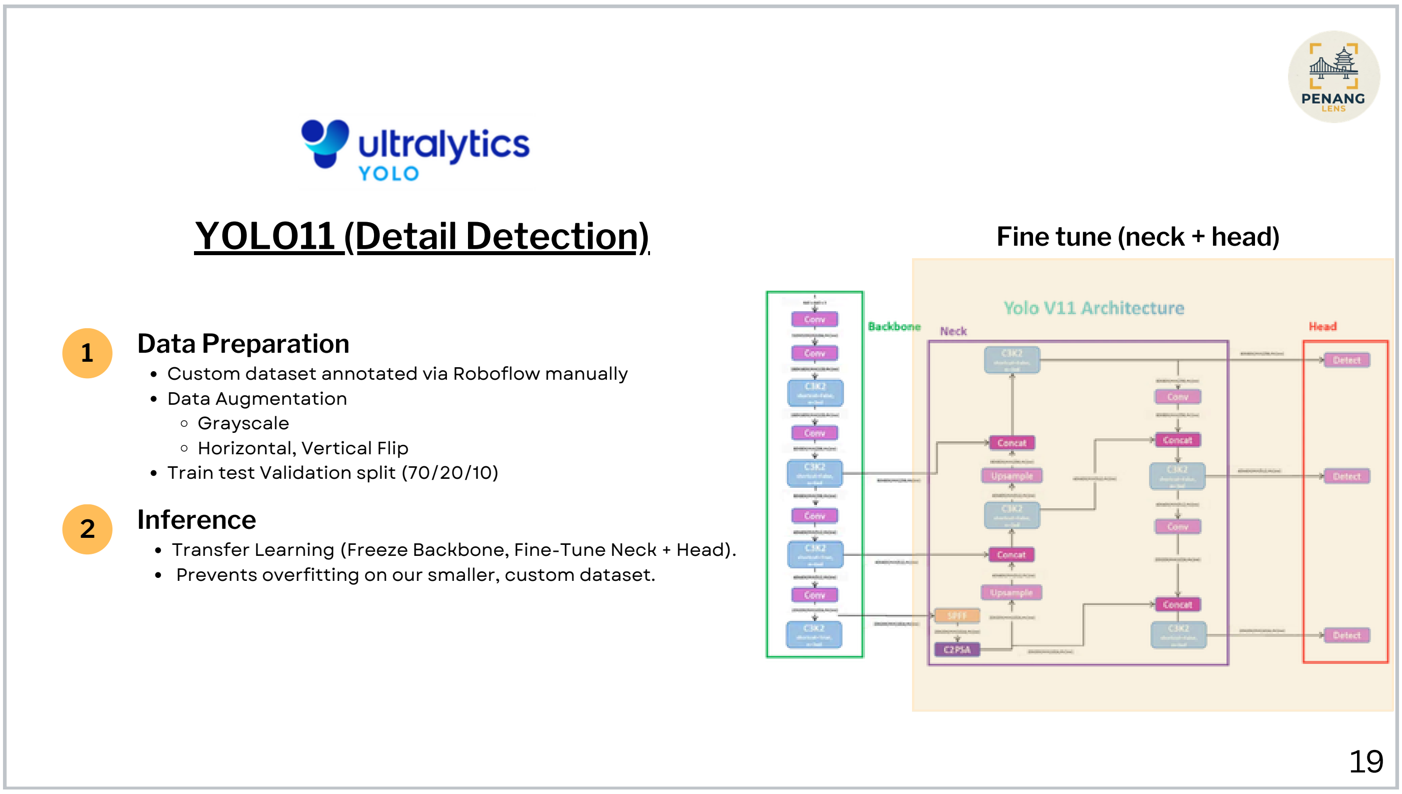
More details
We chose DINOv2, which is a Vision Transformer, specifically because it moves past the

limitations of older CNN-based models. While a CNN looks at images locally, pixel-by-pixel, the Transformer architecture in DINOv2 uses 'global attention.' This allows it to capture the entire geometric 'fingerprint' of a Penang landmark at once, making it far more accurate than traditional models or even text-heavy tools like CLIP when identifying architecture from different angles.



Then for detail detection, we use YOLO11, which is used to detect detailed architectural elements within the image.

At here, data preparation is manual. We curate a custom dataset and annotate them using Roboflow, drawing bounding boxes manually to establish ground truth. Then, data augmentation such as grayscale and horizontal or vertical flips is applied to prepare model against real-world conditions, such as different lighting and angle. The data is then split with a train/test/validation ratio of 70/20/10.



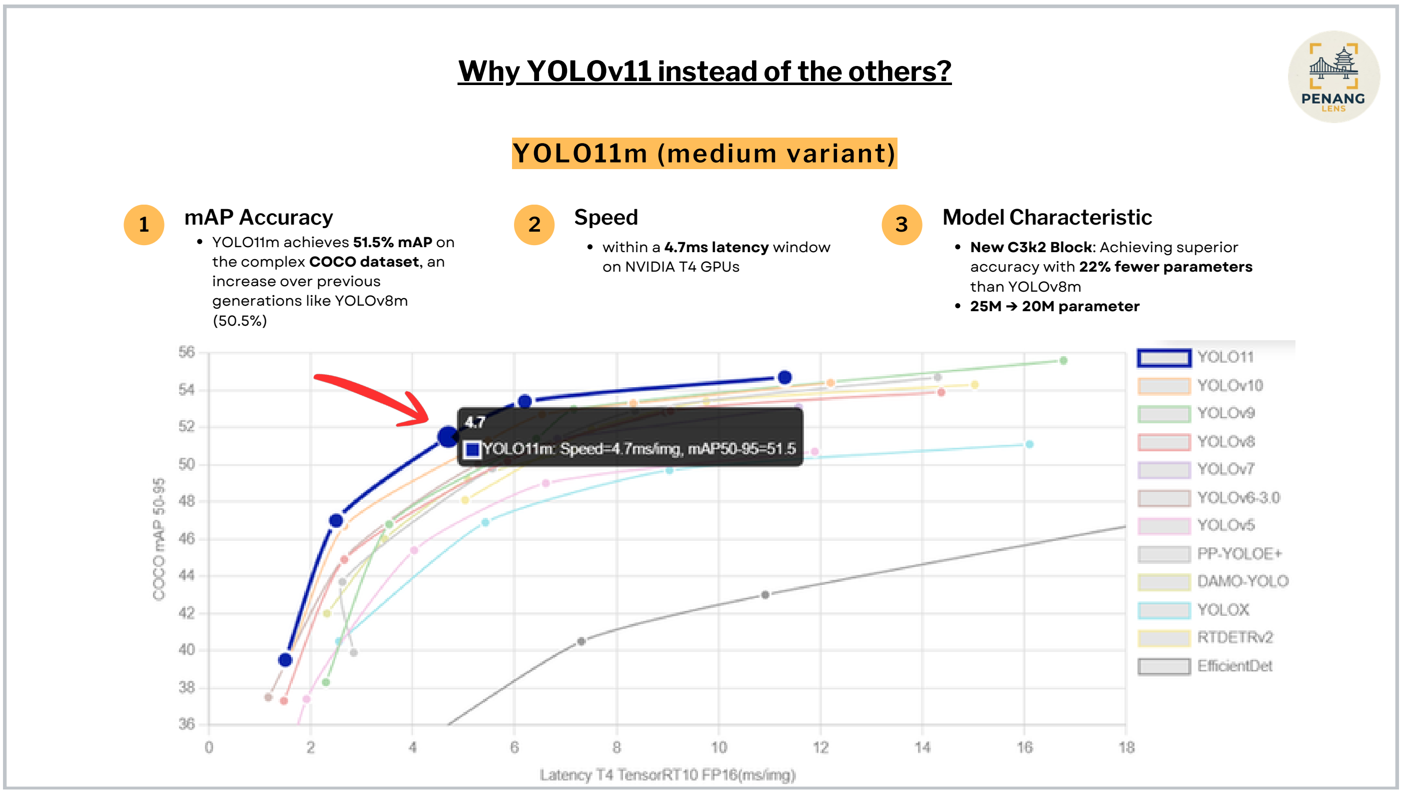
For inference, we apply a transfer learning strategy. We freeze the backbone layers and fine-tune only the Neck and Head using our custom dataset. This is to preserve the foundational knowledge of the model capability like identifying basic visual pattern and to bridge the domain gap with out datasets without overfitting

OTHERS

Backbone: The backbone functions as the primary feature extractor, utilizing a frozen pre-trained network to identify foundational visual patterns such as edges and textures while preventing catastrophic forgetting of general knowledge.

Neck: The neck serves as the feature aggregator, employing the new C3k2 block and C2PSA spatial attention to fuse multi-scale features and isolate the complex geometric nuances of Penang's heritage architecture. (bridge of feature extraction and the prediction, combining fine textures and broad shapes to ensure the model accurately identify architectural details regardless of size n distance)

Head: The head acts as the prediction module, performing the final regression and classification to output high-precision bounding boxes and labels for specific architectural details.



We are using YOLO11, specifically the YOLO11m (medium) variant with 20 million parameters.

First, it offers excellent Accuracy: as shown by the blue line on the graph, YOLO11m achieves nearly 52% mAP on the COCO dataset.

Next is its Speed: it operates within a 4.7ms latency window on an NVIDIA T4 GPU.

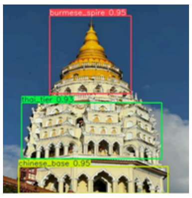
Finally, it provides Enhanced Efficiency: the new C3k2 Block architecture allows YOLO11 to reach this accuracy with 22% fewer parameters. By reducing the parameter count from 25M to 20M, we significantly lower our cloud processing costs.

others
t utilizes a Cross-Stage Partial (CSP) Network design with two smaller sub-blocks, allowing it to capture intricate spatial features and overlapping geometries much more efficiently than previous versions

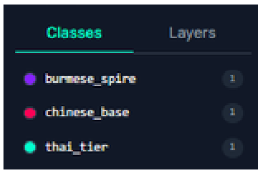
11 landmarks will be targeted in this project including

1) Kapitan Keling mosque
2) Guan Yin Statue
3) Fort cornwallis (cannon and statue) x2
4) Khoo Kong Si
5) Penang Hill funicular train
6) Queen Victoria Memorial Clock Tower
7) city hall esplanade
8) St. George's Church
9) Penang street art x2

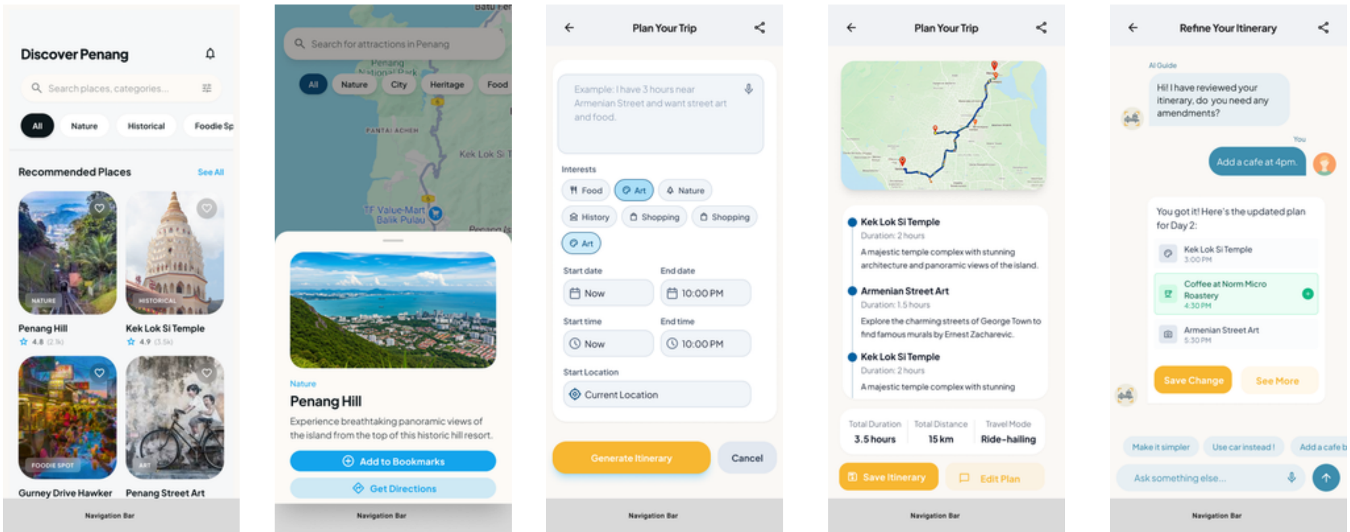
10 manual labelled image each



2 - 6 class in each image



MODULE 3 - INTELLIGENT ITINERARY & TRAVEL COMPANION



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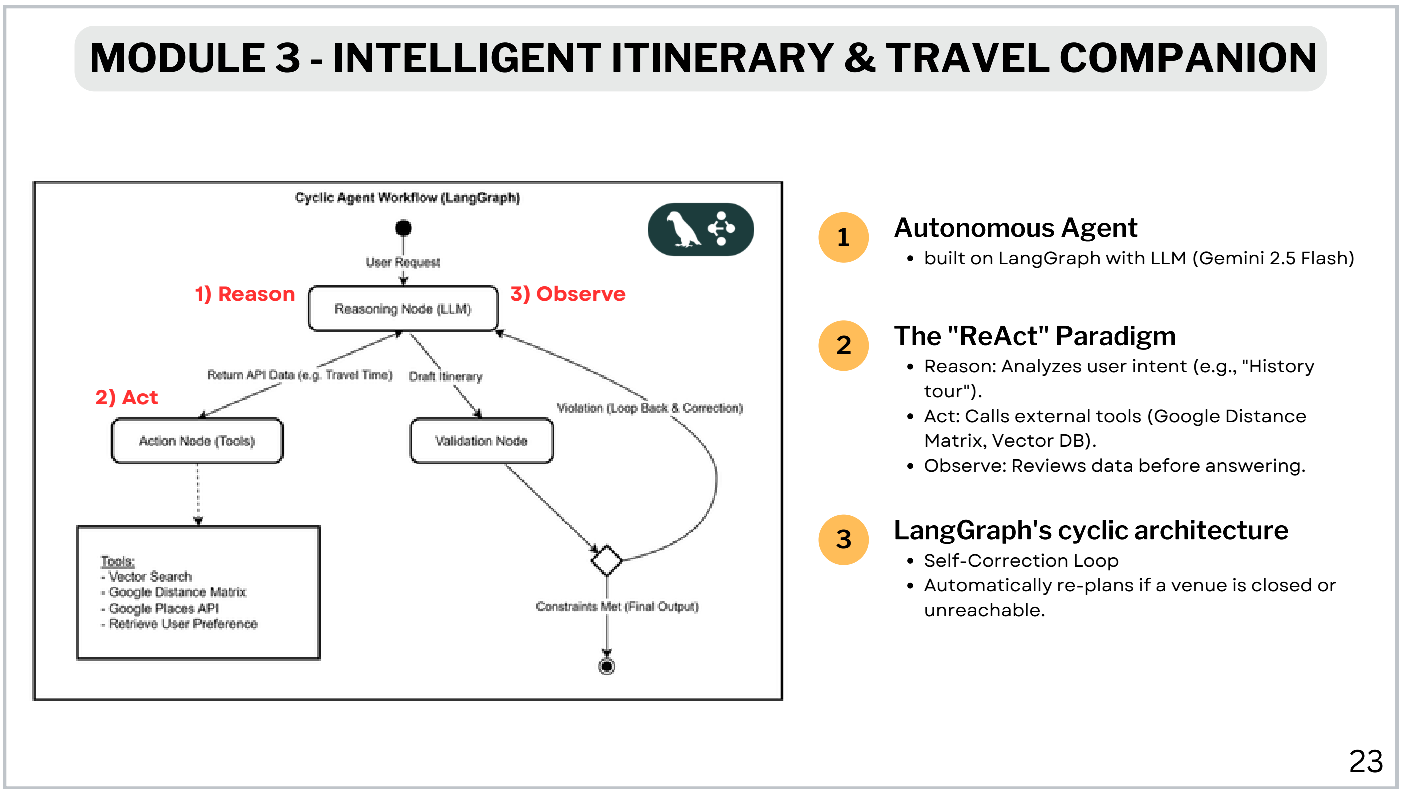
Next move on to module 3, the intelligent itinearrary and travel companion

first the user get to see the reccomended places to visit, thanks to the personalisation engine in module 1

besides, there is also an ai agent chat, where user can insert some inputs and filters and the ai will generate itineraries based on user preference, then user can continue to refined the itinerary in a chat



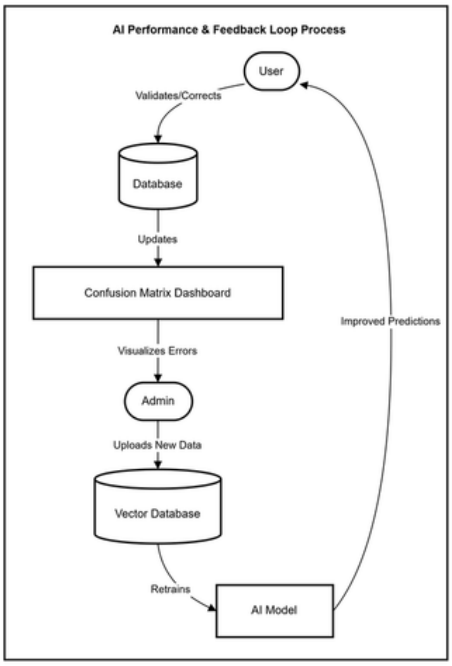
At here, we didn't want to build just a basic chatbot. Instead, we are building an autonomous AI agent orchestrated by LangGraph.

Unlike simple chatbots that simply output text, our agent operates on the ReAct Paradigm —Reason, Act, and Observe. When a user makes a request, the agent first Reasons about what is needed. Then, it Acts by calling real external tools—like the Google Place API to know the operation hours of a shop.

By using LangGraph's cyclic architecture, we can enable a robust Self-Correction Loop.

For example, if the agent finds that a museum is closed, it doesn't just fail or hallucinate. Because LangGraph allows the workflow to loop back, and automatically re-plans the itinerary. It iterates until it finds a feasible alternative, ensuring output is correct.

MODULE 4 - SMART CONTENT CURATION & MANAGEMENT



1

Hierarchical Content Management

- Admin manages complex relationships
  - (e.g., **Parent**: Fort Cornwallis → (**Child**: Francis Light Statue).
- Allows granular creation of spots and sub-POIs.

There are many Point of Interest (POI) at a single Fort Cornwallis

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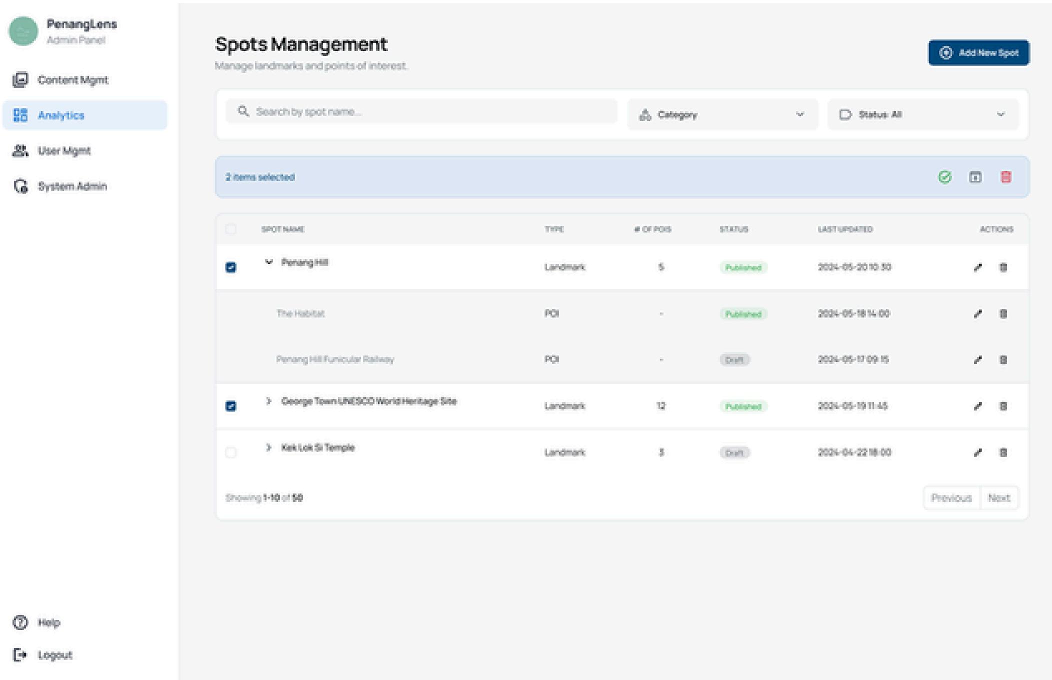
Finally, Module 4 covers the administration and maintenance of the system.
First, we handle Hierarchical Content Management. As u can see in this poster, the torusims attraction Fort cornwallis actually consist of many point of interest.

MODULE 4 - SMART CONTENT CURATION & MANAGEMENT



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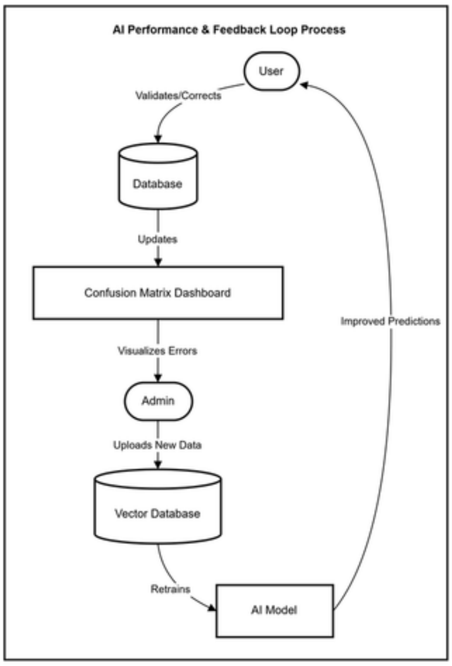
Hence the admin portal shoul allow the Administrators to create landmarks with complex structures, defining 'parent children relationship.

MODULE 4 - SMART CONTENT CURATION & MANAGEMENT



1 **Hierarchical Content Management**

- Admin manages complex relationships
  - (e.g., **Parent**: Fort Cornwallis → (**Child**: Francis Light Statue).
- Allows granular creation of spots and sub-POIs.

2 **Automated AI Indexing**

- Upon image & descriptions upload, the system triggers a background Indexing process
- Automatically extracts and indexes vector embeddings

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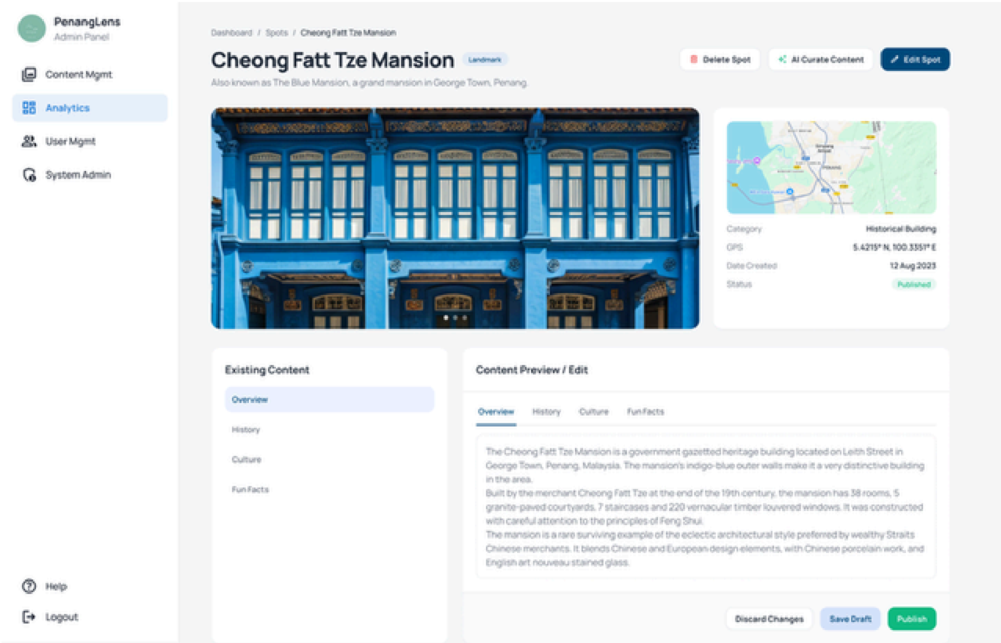
Second, we implemented Automated AI Indexing. When an admin uploads a reference photo for these spots, the system automatically triggers DINOv2 in the background to embed and store the image automatically.

MODULE 4 - SMART CONTENT CURATION & MANAGEMENT



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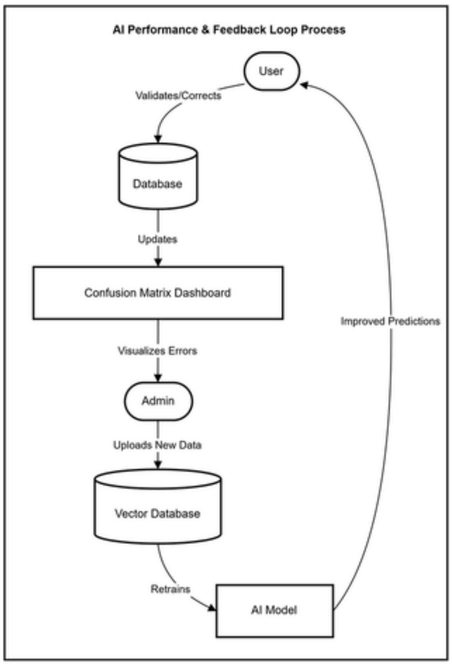
MODULE 4 - SMART CONTENT CURATION & MANAGEMENT



1 Hierarchical Content Management

- Admin manages complex relationships
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- Allows granular creation of spots and sub-POIs.

2 Automated AI Indexing

- Upon image & descriptions upload, the system triggers a background Indexing process
- Automatically extracts and indexes vector embeddings

3 Human-in-the-Loop (Feedback & Evaluation)

- Admin reviews user ratings and feedback on AI-generated stories
- Identifies "Hallucinations" or poor detections to iteratively improve the dataset.

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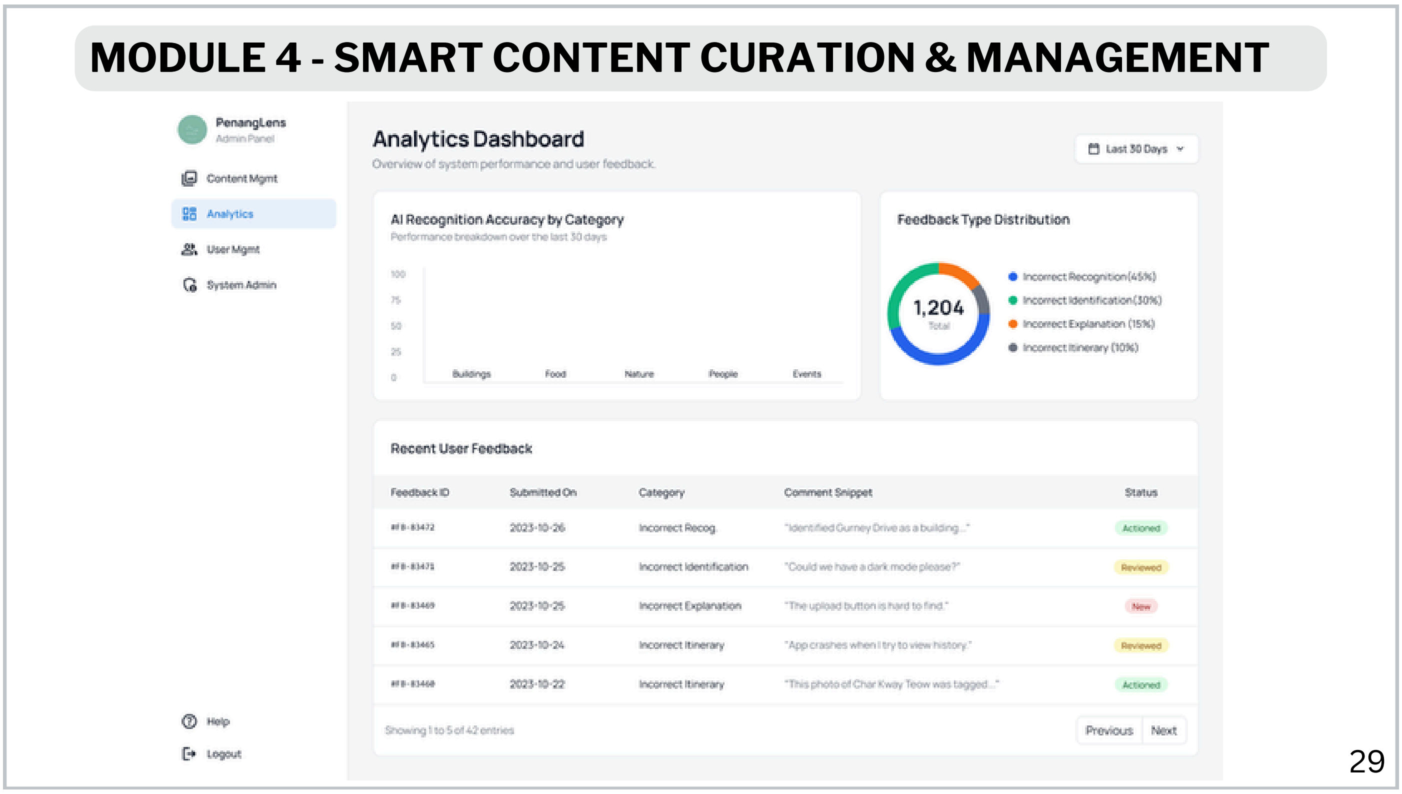
Third, we implement a Human-in-the-Loop Feedback mechanism where Administrators can review user feedback and ratings on the AI-generated stories or AI Visual recognition.



This allows them to spot inaccuracies or 'hallucinations' and correct the database, hence creating a continuous improvement loop.

AI COMPONENTS

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Embedding Model

Gemini Embedding

Google gemini-embedding-001

1. **Matryoshka Representation Learning**
   - This allows the 768-D vectors to be truncated for faster processing without losing much semantic accuracy.

MRL Dimension	MTEB Score
2048	68.16
1536	68.17
768	67.99
512	67.55
256	66.19
128	63.31

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Now lets talk about the embedding model we will be using which is Google gemini-embedding-001, first, it has Matryoshka Representation Learning. This means during the training, the most important information is packed at the start of the vector. As seen in our benchmarks, we can shrink a 768-D vector to 256-D with only a 1% accuracy drop, but in return we got a better processing speed

Embedding Model

PENANG LENS

Gemini Embedding

Google gemini-embedding-001

1. **Matryoshka Representation Learning**
   - This allows the 768-D vectors to be truncated for faster processing without losing much semantic accuracy.
2. **Intent-Aware Task Parameters**
   - Utilizes specific task_type settings (like RETRIEVAL_QUERY or SEMANTIC_SIMILARITY) for different embedding task

Task type	Description	Examples
SEMANTIC_SIMILARITY	Embeddings optimized to assess text similarity.	Recommendation systems, duplicate detection
CLASSIFICATION	Embeddings optimized to classify texts according to preset labels.	Sentiment analysis, spam detection
CLUSTERING	Embeddings optimized to cluster texts based on their similarities.	Document organization, market research, anomaly detection
RETRIEVAL_DOCUMENT	Embeddings optimized for document search.	Indexing articles, books, or web pages for search.
RETRIEVAL_QUERY	Embeddings optimized for general search queries. Use RETRIEVAL_QUERY for queries; RETRIEVAL_DOCUMENT for documents to be retrieved.	Custom search
CODE_RETRIEVAL_QUERY	Embeddings optimized for retrieval of code blocks based on natural language queries. Use CODE_RETRIEVAL_QUERY for queries; RETRIEVAL_DOCUMENT for code blocks to be retrieved.	Code suggestions and search
QUESTION_ANSWERING	Embeddings for questions in a question-answering system, optimized for finding documents that answer the question. Use QUESTION_ANSWERING for questions; RETRIEVAL_DOCUMENT for documents to be retrieved.	Chatbox
FACT_VERIFICATION	Embeddings for statements that need to be verified, optimized for retrieving documents that contain evidence supporting or refuting the statement. Use FACT_VERIFICATION for the target text; RETRIEVAL_DOCUMENT for documents to be retrieved	Automated fact-checking systems

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Beside there is also intent aware task parameters in this model, different task type such as retrieval query or semantic similarity, so we get to use different mathematical calculation that suits our use case.

Detail version: (not used)

Next, we utilize Intent-Aware Task Parameters to optimize the mathematical focus for each specific use case. For the Personalization task, we apply SEMANTIC_SIMILARITY to symmetrically match user interest tags with landmark descriptions. For our AI Vision pipeline, we use RETRIEVAL_QUERY for the detected visual labels and RETRIEVAL_DOCUMENT for our heritage database to bridge the gap between short search phrases and deep historical context

Embedding Model

Gemini Embedding

Google gemini-embedding-001

1. **Matryoshka Representation Learning**
   - This allows the 768-D vectors to be truncated for faster processing without losing much semantic accuracy.
2. **Intent-Aware Task Parameters**
   - Utilizes specific task_type settings (like RETRIEVAL_QUERY or SEMANTIC_SIMILARITY) for different embedding task
3. **Heritage-Specific Cultural Intelligence**
   - High performance in MTEB multilingual benchmarks and Semantic Textual Similarity (STS) ensures precise mapping of complex, multi-ethnic heritage terminology within the vector space.

BENCHMARK	MODEL	gemini-embedding-001
MTEB (Multilingual)	Mean (Task)	68.37
	Mean (TaskType)	59.59
	- Bitext Mining	79.28
	- Classification	71.82
	- Clustering	54.59
	- Inst. Retrieval	5.18
	- Multilabel Class.	29.16
	- Pair Class.	83.63
	- Reranking	65.58
	- Retrieval	67.71
	- STS	79.4
MTEB (Eng, v2)		73.3
MTEB (Code, v1)		76
XOR-Retrieve		90.42
XTREME-UP		64.33

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Finally, the model is suitable for our tourism and heritage use case. As it is trained with large global corpus and a high 79.4 STS score, it understands the semantic relationship between niche terms in Penang tourism.

Vector Search Engine & Storage

Azure AI Search

AI Search

Component	Personalisation & RAG	AI Vision (DINOv2)
Search Engine Input	Text + (256 & 768-d vector)	(768-d vector only)
API Payload	**Hybrid Search**: Combines vector and keyword	**Pure Vector Search**: Visual similarity
Search Algorithm	HNSW (Vector) + BM25 (Keyword)	HNSW (Hierarchical Graph Search)
Ranking Strategy	Reciprocal Rank Fusion (fusing k-NN + BM25 results)	k-NN (Nearest Neighbor Selection)

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Now lets focus on the Azure AI Search which is our search engine and storage

For Personalization and RAG, we perform a Hybrid Search where we will send a payload containing both the vector and the keyword. The system uses the HNSW algorithm to perform a fast k-NN search for semantic matches, while BM25 ranks for keyword similarity. Finally, Reciprocal Rank Fusion, mathematically fuses these two separate ranked lists into a single, optimized result.

In contrast, our AI Vision pipeline uses a Pure Vector Search. So similarly we use the HNSW algorithm to perform a k-NN search, identifying the most similar image,

Key Performance Metrics & AI Evaluation

1 **Landmark Identification (DINOv2 + Vector Search)**

- Metric: Top-1 Accuracy.
- **Target: > 85%**.
  - *Measures how frequently the correct Landmark ID is retrieved as the primary match.*

2 **Detail Detection (YOLO11)**

- Metric: Mean Average Precision (mAP@0.5).
- **Target: > 0.75.**
  - *Measures the precision and bounding box accuracy for specific architectural features (e.g., Roofs, Carvings).*

3 **Generative AI & RAG Evaluation (Ragas Framework)**

- **Faithfulness (Hallucination Check):**
  - Target: > 0.9
    - *Ensures all generated stories are strictly grounded in the retrieved historical database context.*
- **Answer Relevance:**
  - Target: > 0.85.
    - *Evaluates how effectively the AI Agent's itinerary and stories address specific user constraints (e.g., "History focus").*

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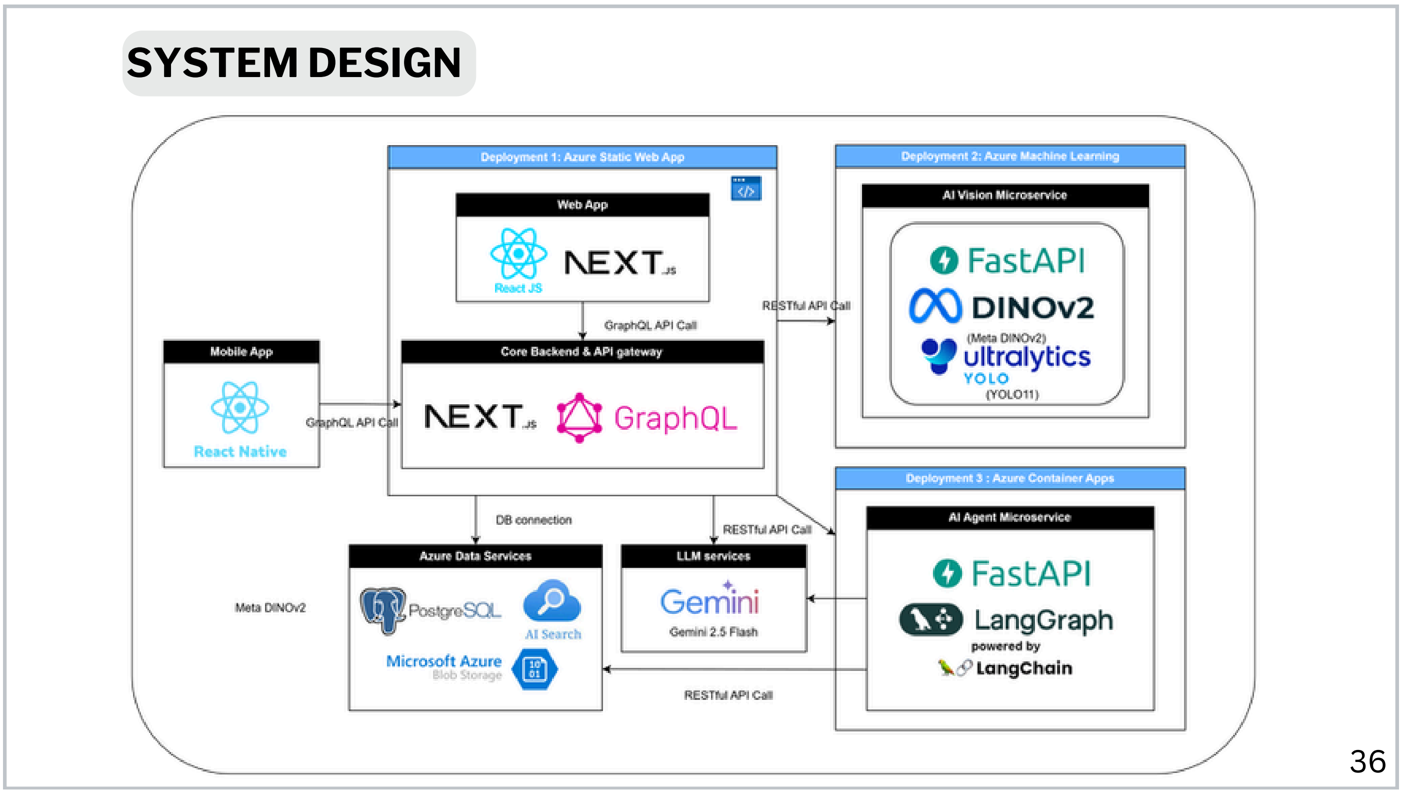
Now lets talk about how we evaluate our ai model.

First, for our landmark identification of DINOv2, it shall have a 85% accuracy in identifying the correct landmark.

For Detail detection of yolo11, it shall have a 0.75 mean average precision ensuring we accurately draw the bounding boxes on photos.

Last, we validate our Gen AI using the Ragas Framework, which uses a 'Judge LLM' to evaluate the result. It shall target a 0.9 of faithfulness to pass the hallucination check and a 0.85 answer relevancy to ensure response match the user preference.



This is the system design of PenangLens, which follows a Microservice Architecture

Three deployments will be made , to handle the backend, ai vision microservices and ai agent microservices.

This modular microservice architecture allows the project to be scalable in the future.

Finally, To conclude, PenangLens is a Unified Intelligent Platform that combines visual discovery, storytelling, and planning into one seamless experience.

Thank you, and I am now ready for your questions.

Q&A

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THANK YOU

APPENDIX

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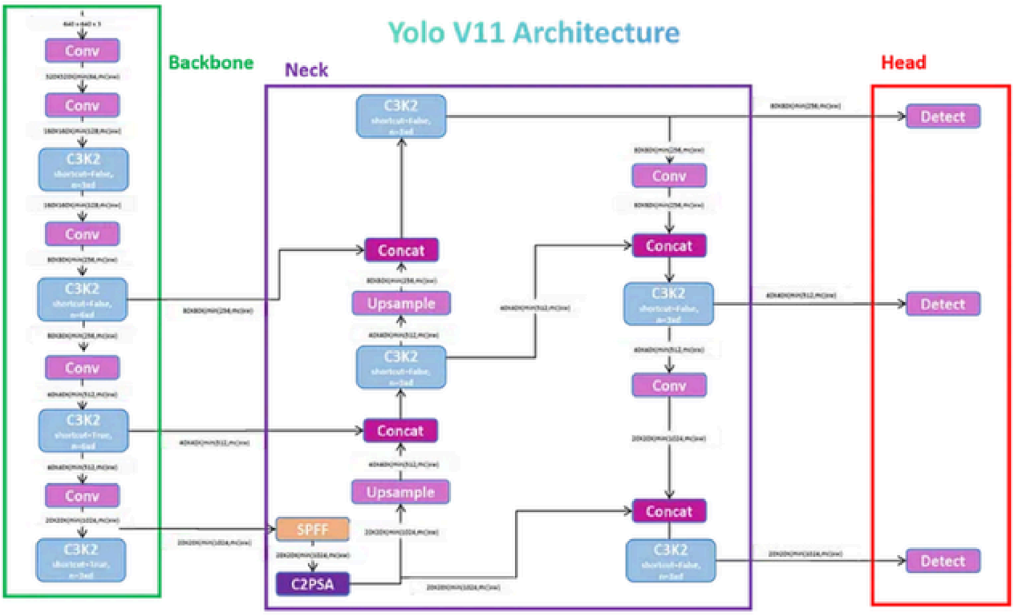
YOLO V11 ARCHITECTURE



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DINO ALTERNATIVE?

Model	Vector Type	What the Vector Represents	How it Learns
ResNet (Supervised CNN)	**Label-Driven Vectors**	These vectors represent "probabilities" that an image belongs to a specific human-defined category.	Learns by being told "This is a temple" thousands of times.
CLIP (Contrastive)	**Semantic-Textual Vectors**	These vectors represent the "description" of an image, aligning visual data with natural language.	Learns by matching images to captions (e.g., "A photo of a colonial building").
DINOv2 (Self-Supervised)	**Structural Fingerprints**	These vectors represent the unique **geometric and visual essence** of the object itself, regardless of its name.	Learns by comparing different views of the same image to find consistent features.

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while ResNet and CLIP provide useful vectors for general classification and text-based search, they fail to provide the granular, instance-level identification required for PenangLens. DINOv2 acts as a "Visual Information Retrieval" specialist

In DINO, the student network learns by adjusting its output to match the teacher network’s output for different augmented views of the same image.

DINO VERSION?

DINOv3 performance benchmarks

Task Benchmark	DINO ViT-B/8 0.09B	DINOv2 ViT-g/14 1.1B	DINOv3 ViT-7B/16 7B	SigLIP 2 ViT-g-OPT/16 1.8B	PE ViT-G/14 1.9B
Segmentation ADE-20k	31.8	49.5	**55.9**	42.7	38.9
Depth estimation NYU ↓	0.537	0.372	**0.309**	0.494	0.436
Video tracking DAVIS	68.7	76.6	**83.3**	62.9	49.8
Instance retrieval Met	17.1	44.6	**55.4**	13.9	10.6
Image classification ImageNet ReaL	85.9	89.9	90.4	**90.5**	90.4
ObjectNet	39.9	66.4	79.0	78.6	**80.2**
Fine-grain image classification iNaturalist 2021	68.3	86.1	**89.8**	82.7	87.0

Metric	DINOv2 (ViT-B/14)	DINOv3 (ViT-7B/16)
Parameters	**~86 Million**	**~6.7 Billion**
Embedding Size	**768-d**	**4096-d**
VRAM Needed	**< 2 GB**	**14–20 GB (Min)**
Ideal Hardware	**Standard GPU**	**Enterprise A100/H100**
Cloud Cost	**Low (Budget-friendly)**	**Very High**

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Supervised Convolutional Neural Networks (CNNs): Traditional models like ResNet have been used for years to extract features. However, they often rely heavily on large, manually labeled datasets, which limits their effectiveness for specific, under-documented heritage sites.

Swin Transformers: These are transformer-based architectures that have been shown to provide more robust feature representations than standard CNNs in tasks like heritage building analysis.

Image-Text Embedding Models (e.g., CLIP): These models, used in related research projects like MACAVA, align images and text in the same vector space. This allows a system to search for images using text queries, though it may be less precise for exact landmark matching than DINOv2.

Google Lens: This is a commercial-grade tool that uses deep learning for visual lookup. While it identifies landmarks accurately, it often creates a "Semantic Gap" by providing a label without the deep architectural or narrative context required for a specialized application like PenangLens.

YOLO-World: An open-vocabulary detection model that can identify objects it was not specifically trained on. While flexible, it is generally less precise and slower for fine-grained identification compared to a specialized pipeline using DINOv2 and YOLO11.

COMPARISON MATRIX

Feature	PenangLens	Layla AI	Google Lens	TripAdvisor	Audorista
AI Landmark Recognition	✅	❌	✅	❌	❌
Contextual Detail Recognition	✅	❌	❌	❌	❌
Dynamic Story Generation	✅	❌	❌	❌	❌
Interactive AI Chat	✅	✅	❌	❌	❌
Specialized Heritage Focus	✅	❌	❌	✅	✅
Personalization Engine	✅	✅	❌	❌	❌
AI Itinerary Planning	✅	✅	❌	❌	❌

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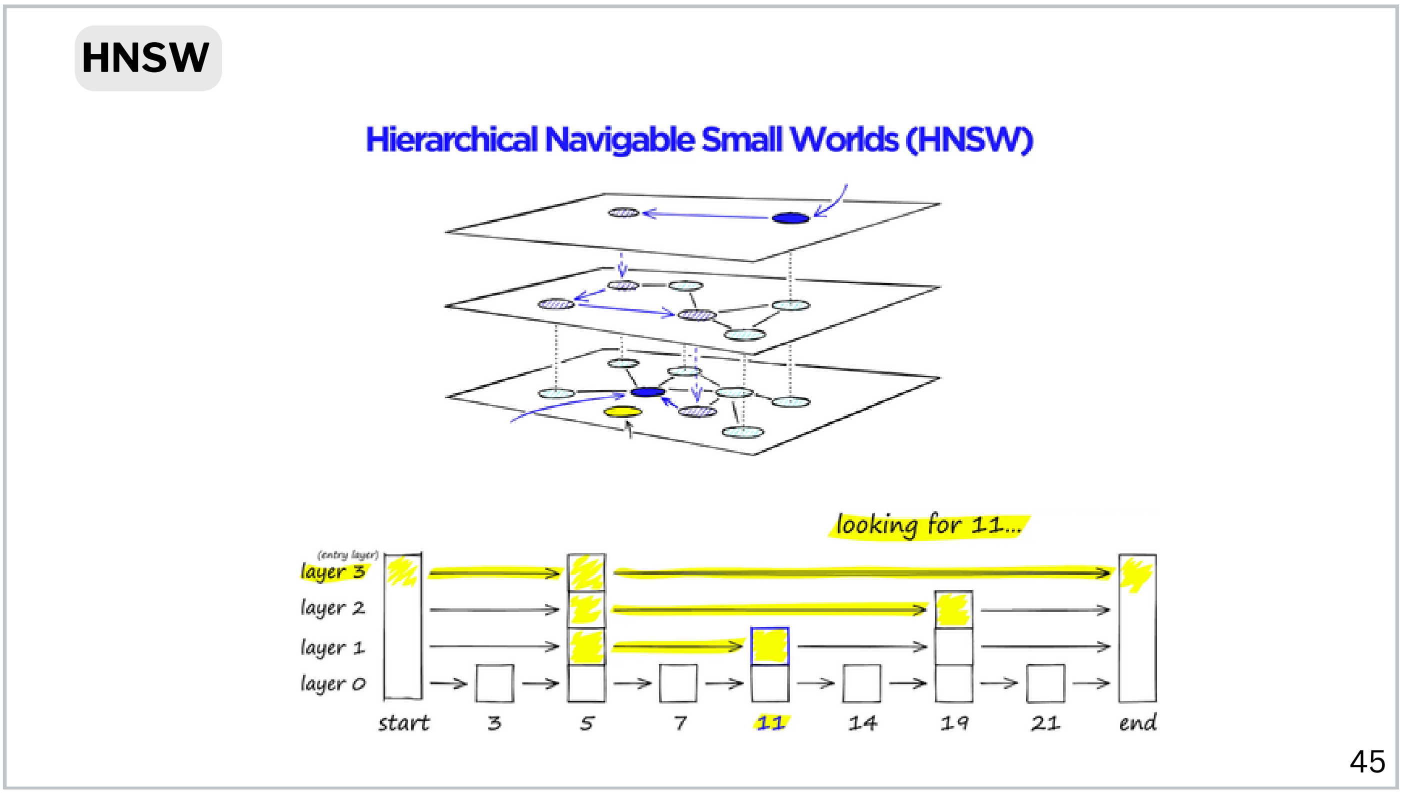

Definition: A graph-based indexing algorithm designed for high-performance Approximate Nearest Neighbor (ANN) retrieval.

Mechanism: It creates a multi-layered graph where the top layers contain "expressway" connections to jump across data and the bottom layers contain fine-grained local connections.

HNSW is a graph-based approximate nearest neighbor algorithm that finds similar vectors quickly by navigating a multi-layer network, trading a small amount of accuracy for very fast search at scale.

BM25

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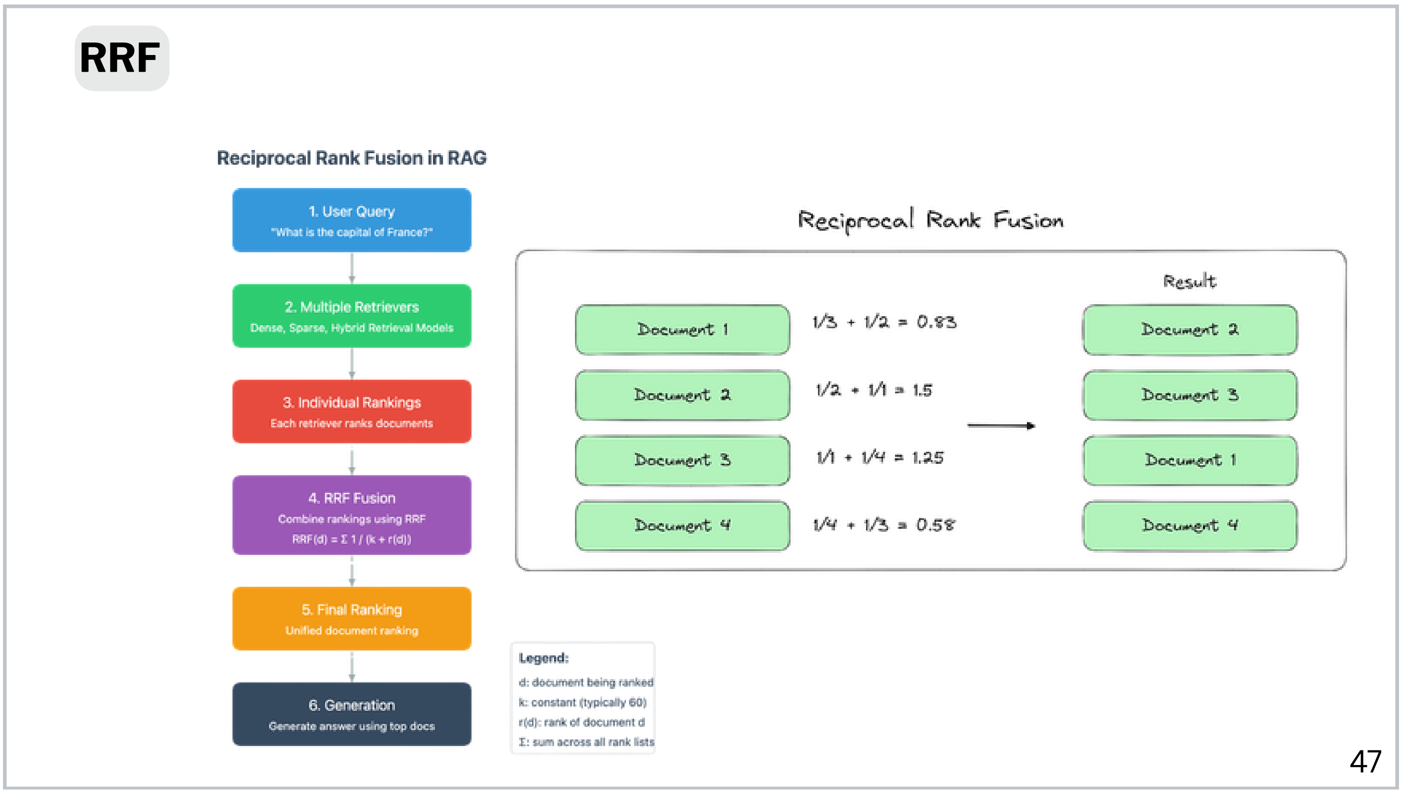
BM25 works by analyzing the intersection between documents that simply contain our search term and those that are statistically relevant to the query. It calculates a relevance score based on how unique a keyword is. For example, a rare word like 'Peranakan' will carry much more weight than a common word like 'building'.

In our system, this allows us to filter out the 'noise' and find only the specific subset of Relevant documents containing the query term. This ranking is then passed to our RRF process, where it is fused with our vector search results to give the tourist the single most accurate historical fact available.

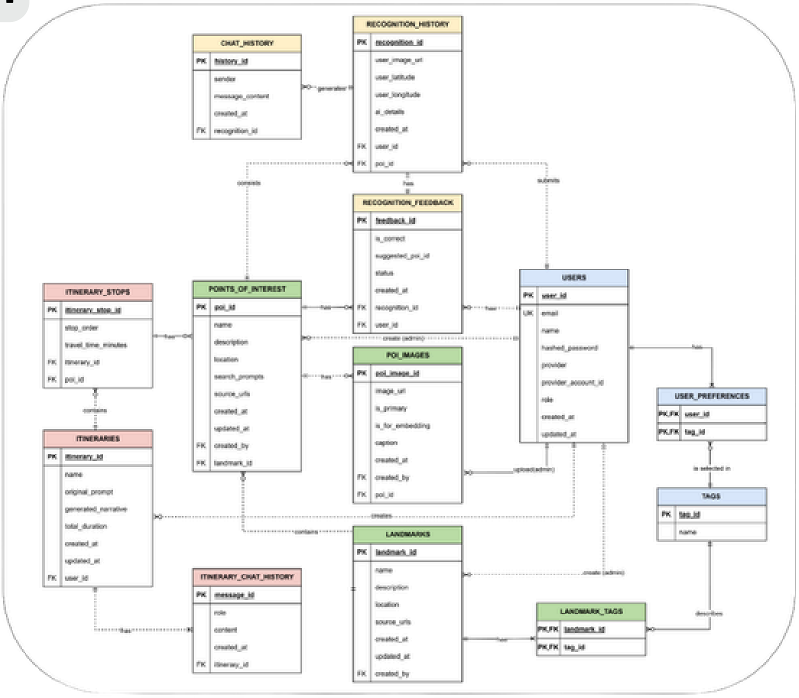


Now lets talk about the decision maker in our architecture, which is RRF. As you can see from the formula, RRF calculates a score by taking the reciprocal of a document's rank in our separate search lists.

The logic is simple: it rewards agreement. If our HNSW vector search says a document is relevant based on meaning, and our BM25 search says it is relevant based on keywords, RRF gives it a massive boost.

By using this calculation, we solve the common problem where an AI might find something that looks right but has the wrong name, or something with the right name but the wrong context. RRF ensures that the final result passed to our Gemini LLM is the one that both 'scouts' agree is the most accurate historical fact for the tourist.

ERD DIAGRAM



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